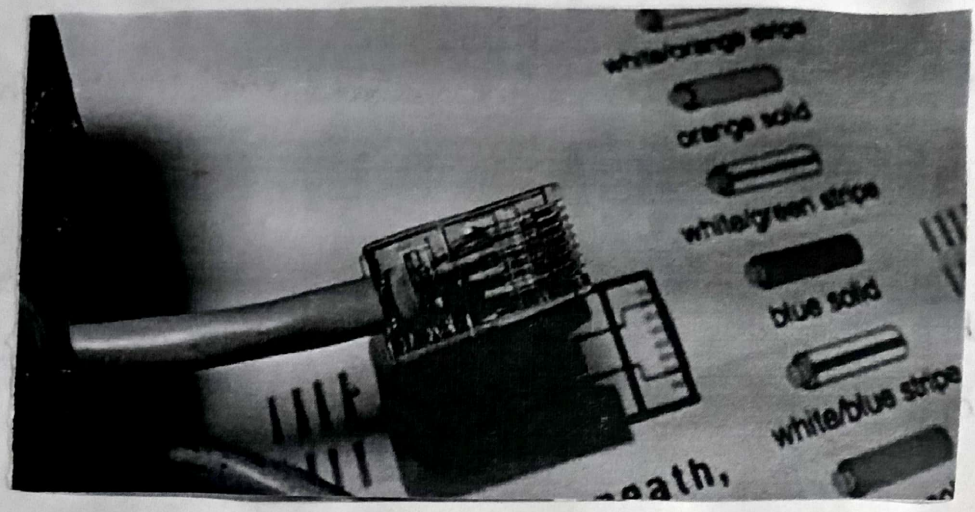
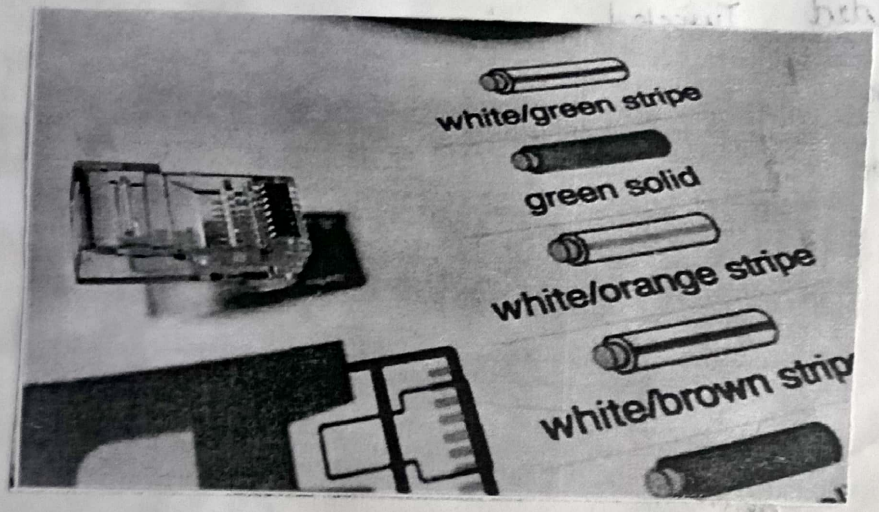
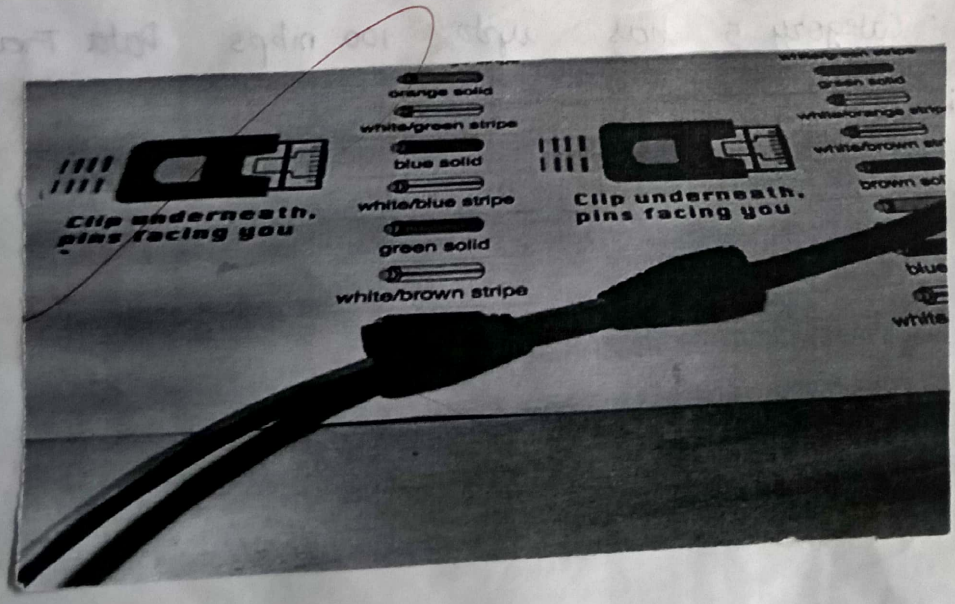






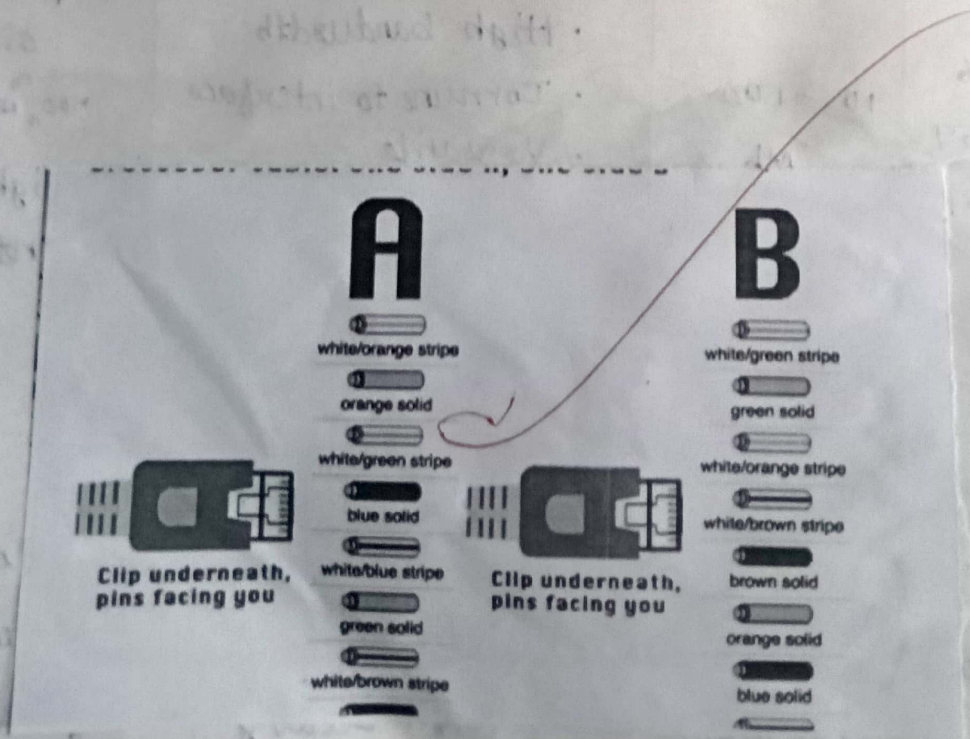
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Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Applications
UTP	Category 3	10 bps	Advantages • Cheaper in cost • Easy to install as they have small overall diameter Disadvantages • More prone to EMI	10 Base-T ethernet
	Category 5	Upto 100 mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages • shielded • Faster than UTP • Less prone to noise Disadvantages • Expensive • Greater installation process	• Gigabit ethernet • Widely used in data centres
SSTP	Category 7	10 Gbps		• Gigabit Ethernet, 10 G Ethernet
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages • High bandwidth • Immune to interface • Versatile Disadvantage • Cost • Size is bulky	• Speed of signal is 500 m • High speed internet connection
Fibre optics cable	Singlemode Multimode	100 Gbps	Advantages • High speed • High bandwidth • High security Disadvantage • expensive	• Maximum distance of fibre optics cable is around 100 miles

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Make your

Tools and

supports.

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Make your own Ethernet Cross-over cable / Straight Cable:

Tools and parts needed:

- Ether cabling. Cat 5e is certified for gigabit. support. but .Cat 5. cabling works as well
- A crimping tool
- Two RJ45 plugs
- Optional two plug shields

Step 1:

To start construction of the device, begin by threading shields on to the cable

Step 2:

Next strip approximately 1.5 cm of the cable shielding from both ends

Step 3:

After you will need to untangle the wires; there should be four twisted pairs.

Step 4:

Once the order is correct, bunch them to gather in a line and if there are any that stick out further than others snip them back to create an even level. The difficult aspect is placing the RJ45 plug without mixing up the order.

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